Spectral and Bispectral Neutron Multiplicity Counting for Plutonium Mass Assay

A simulation study with OpenMC

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Abstract

Passive neutron multiplicity counting finds the mass of a plutonium item from the time correlations between detected neutrons. The standard instrument is the shift register, which counts neutrons in time gates and needs calibrated gate fractions f_d and f_t. This report tests a frequency-domain alternative. Doubles are read from the 3-bank *cross power spectral density* and Triples from the 3-bank *cross-bispectrum*. In these spectra the uncorrelated background cancels by construction, and no gate fractions are needed. We simulate a 300 g Pu metal sphere in a 3-bank He-3 well counter with OpenMC, build 300 s list-mode pulse trains, and analyse them with both methods. Over repeated measurements the bispectrum is unbiased ($+0.8 \pm 1.1\%$ on the mass) but noisier (≈ 9 g per 300 s run). A shift register calibrated with a Cf-252 source is precise (≈ 2 g) but biased by $+4.5\%$, and by $+5.3\%$ for a moderated item, because the calibration source has no fission-chain timing. The best result comes from a hybrid in which the bispectrum calibrates the shift-register gate fraction: about 3 g RMS error on 300 g.

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1 Neutron multiplicity counting in brief

Pu-240 (and the other even isotopes) fissions spontaneously and emits ν neutrons *at the same time*, with ν distributed by $P(\nu)$. Other neutrons are emitted one at a time by (α, n) reactions on light elements. Some neutrons cause further fissions inside the item (self-multiplication M). A well counter moderates the neutrons in polyethylene and captures them in He-3. The capture of each neutron is delayed by a random die-away time, about $\tau \approx 50\,\mu$s.

The detector therefore sees a stream of pulses with a Poisson background of uncorrelated neutrons, plus small *bursts* of correlated neutrons from each fission chain. Three rates summarise this stream:

- **Singles** S: all detected neutrons per second.
- **Doubles** D_∞: correlated *pairs* per second (pairs from the same fission chain).
- **Triples** T_∞: correlated *triplets* per second.

The Böhnel point model links these three rates to three unknowns: the spontaneous-fission rate F (proportional to the Pu-240 effective mass), M, and $\alpha = (\alpha, n \text{ neutrons})/(\text{SF neutrons})$:

$$S = F\varepsilon M\nu_{s1}(1+\alpha), \tag{1}$$

$$D_\infty = \tfrac{1}{2}F\varepsilon^2M^2\Big[\nu_{s2} + \tfrac{M-1}{\nu_{i1}-1}\nu_{s1}(1+\alpha)\nu_{i2}\Big], \tag{2}$$

$$T_\infty = \tfrac{1}{6}F\varepsilon^3M^3\Big[\nu_{s3} + \tfrac{M-1}{\nu_{i1}-1}\big(3\nu_{s2}\nu_{i2} + \nu_{s1}(1+\alpha)\nu_{i3}\big) + 3\big(\tfrac{M-1}{\nu_{i1}-1}\big)^2\nu_{s1}(1+\alpha)\nu_{i2}^2\Big]. \tag{3}$$

Here ε is the detection efficiency, and ν_{sk}, ν_{ik} are the factorial moments of the spontaneous and induced fission multiplicity distributions. Eliminating F and α gives a cubic equation in M. Its physical root ($M \geq 1$) gives M, then F, then α. The Pu mass is $m_{\mathrm{Pu}} = F/(473.5\,\mathrm{s^{-1}g^{-1}})/f_{240}$.

The whole problem is measuring D_∞ and T_∞ accurately. This report compares two ways of doing so.

2 Simulation set-up

2.1 What OpenMC is

OpenMC is an open-source Monte Carlo neutron and photon transport code, developed at MIT and Argonne. It tracks individual particles through a 3D constructive-solid-geometry model using continuous-energy nuclear data (here ENDF/B-VIII.1). It supports fixed-source problems. Its `collision_track` feature writes every selected collision to an HDF5 file, with position, time, reaction, nuclide and the history ID. This event-by-event output is what a list-mode multiplicity study needs.

2.2 Model

2.3 OpenMC features used

- `run_mode = 'fixed source'`, `survival_biasing = False` (analog captures), one batch of 400 000 histories.
- `collision_track`: cells 101–103, nuclide He-3, reaction MT 101 (absorption). Each record is one detection, with its `time`, position `r` (which gives the bank) and `parent_id` (the source history).
- `surf_source_write` on the sphere surface: the net outward crossings per source neutron give the leakage multiplication M_{MC}.

Item	Value
Pu sphere	300 g, $\rho = 15.8\,\mathrm{g/cm^3}$, $r = 1.654\,\mathrm{cm}$; 94 wt% Pu-239, 6 wt% Pu-240 ($m_{\mathrm{eff}} = 18\,\mathrm{g}$)
SF rate	$F = 473.5 \times 18 = 8523\,\mathrm{fissions/s}$; $\alpha = 0.50$
Cavity	void cylinder, $r = 4\,\mathrm{cm}$, $\lvert z\rvert < 4\,\mathrm{cm}$ (end plugs close the top and bottom)
Detector body	cylinder $r = 20\,\mathrm{cm}$, $\lvert z\rvert < 20\,\mathrm{cm}$; HDPE ($0.96\,\mathrm{g/cm^3}$, S($\alpha, \beta$) `c_H_in_CH2`)
Absorber	He-3 mixed homogeneously, 1.2×10^{-5} atoms/(b cm)
Banks	3 azimuthal 120° sectors X, Y, Z (cells 101–103)
Source	uniform in the sphere, isotropic, Pu-240 SF Watt spectrum ($a = 0.795\,\mathrm{MeV}$, $b = 4.689\,\mathrm{MeV^{-1}}$)

Table 1: Model parameters (`config/assay_config.json`).

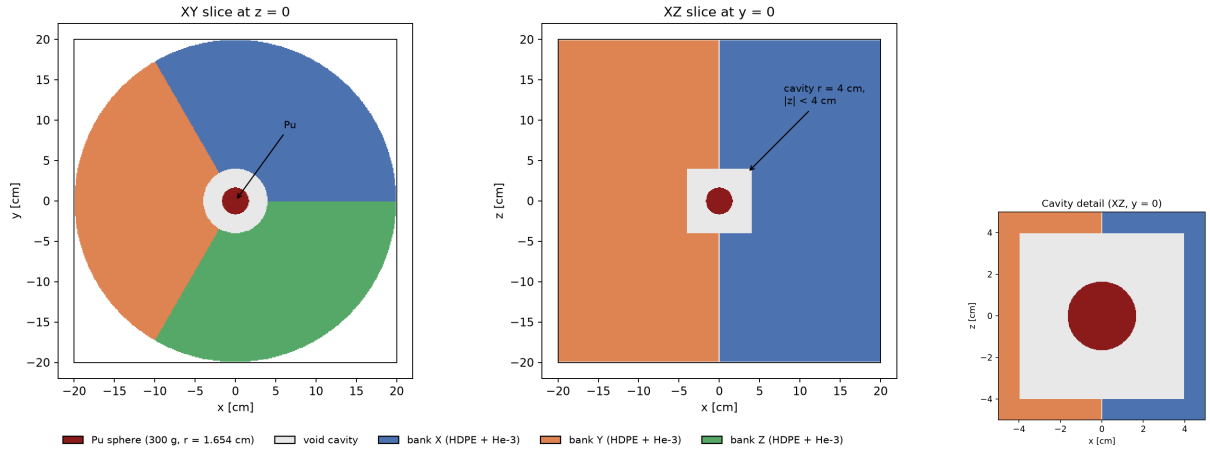


Figure 1: OpenMC geometry, rasterised from the model's own cell definitions. Left: XY slice showing the three 120° detector banks around the cavity. Middle: XZ slice; the end plugs enclose the sphere almost 4π. Right: cavity detail.

- Default physics otherwise. OpenMC treats fission *implicitly* in fixed-source mode: fission neutrons are banked as an expected value at each Pu collision. The mean multiplication is right, but the spread of neutrons per induced fission is not. We keep this as an approximation and calibrate *effective* induced-fission moments $\nu_{i2} = 4.85$, $\nu_{i3} = 0.22$ from the transport output. Both methods use them, so the comparison is fair.

2.4 From single histories to a measurement

OpenMC gives the response to *one* source neutron starting at $t = 0$: a list of (capture delay, bank) pairs. A 300 s measurement is then assembled in Python:

1. Draw $N_{\mathrm{SF}} \sim \mathrm{Poisson}(FT)$ fission times, uniform in $[0, T]$, with $\nu \sim P(\nu)$ (Pu-240).
2. Draw (α, n) emission times at rate $F\nu_{s1}\alpha$, one neutron each.
3. Give every emitted neutron a random OpenMC history, and shift its captures by the birth time.
4. Merge everything and sort by time. The result is 6.85 million pulses with time stamps and bank labels.

Because every step is known, the *exact* expected rates follow from the factorial moments of the per-history detected counts (m_1, m_2, m_3). We use them as ground truth: $S = 22854$, $D_\infty = 17404$, $T_\infty = 13835\,\mathrm{s^{-1}}$.

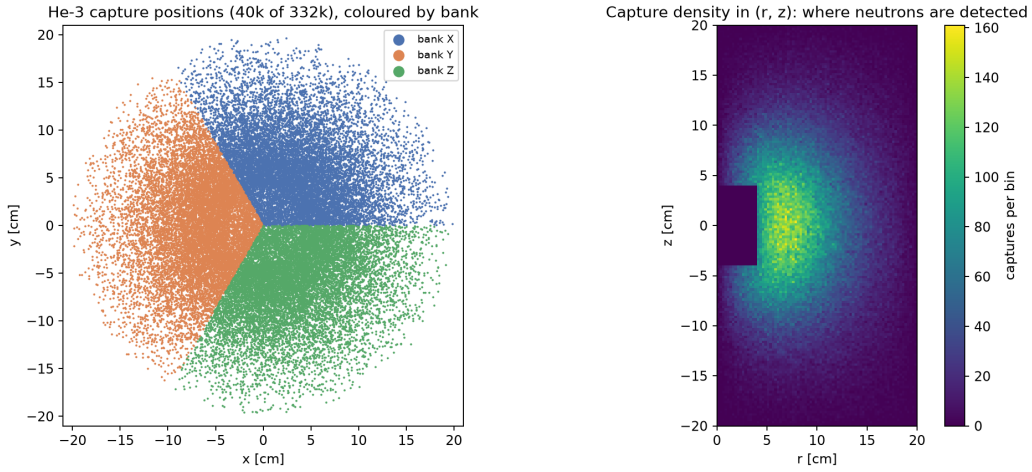


Figure 2: Where neutrons are detected (400k source histories). Left: He-3 capture positions coloured by bank. Right: capture density in (r, z). Most captures occur 3–10 cm into the polyethylene, after thermalisation.

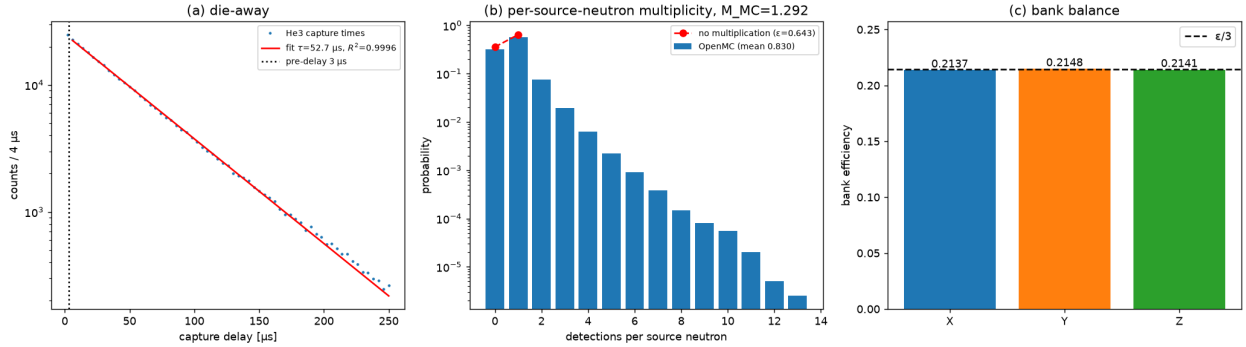


Figure 3: Transport results. (a) He-3 capture-time distribution: $\tau = 52.7\,\mu$s, $R^2 = 0.9996$. (b) Detected counts per source neutron. The tail beyond 1 comes from induced fission ($M_{\mathrm{MC}} = 1.292$). (c) Bank balance: $\varepsilon = 0.643$, banks equal within 0.3%.

3 Legacy method: the shift register

Every pulse opens two gates of width $G = 64\,\mu$s:

- an **R+A** gate just after a pre-delay $P = 3\,\mu$s: real (correlated) + accidental counts;
- an **A** gate after a long delay $L = 2000\,\mu$s, when the correlation has died away: accidentals only (8 gates averaged).

Counting how many pulses fall in each gate gives the histograms $f(n)$ and $b(n)$. Their reduced factorial moments give

$$D = S(f_1 - b_1), \qquad T = S\big[(f_2 - b_2) - b_1(f_1 - b_1)\big]. \tag{4}$$

These are *gated* rates. Only the fraction of correlated neutrons that falls inside the gate is counted:

$$D_\infty = D/f_d, \qquad T_\infty = T/f_t, \qquad f_d = e^{-P/\tau}(1 - e^{-G/\tau}),\ f_t \approx f_d^2. \tag{5}$$

We use empirical $f_d = 0.657$, $f_t = 0.434$, taken from the simulated die-away profile with the neutrons assumed independent. This is the weak point. f_t depends on the detailed time structure of the fission chain, and in practice it comes from a calibration source.

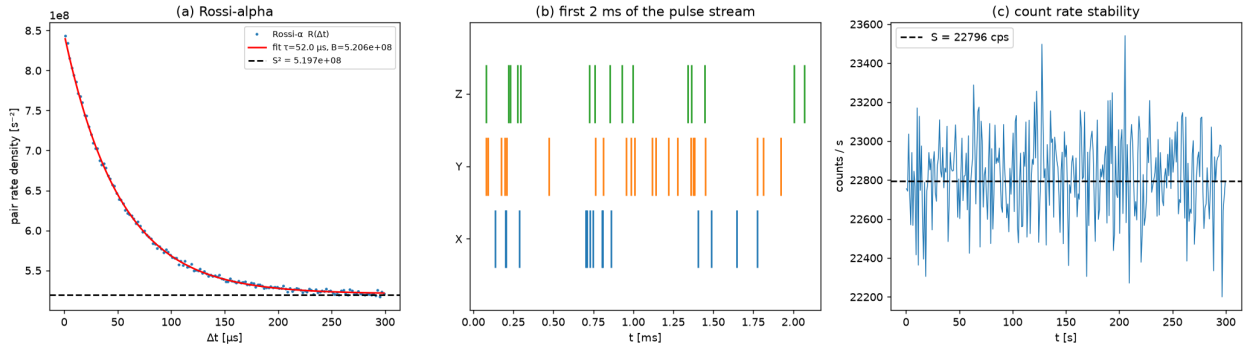


Figure 4: Synthetic pulse train. (a) Rossi-alpha distribution: correlated excess over the S^2 baseline, decaying with $\tau \approx 52\,\mu$s. (b) First 2 ms of the stream per bank. (c) Count rate is stable over 300 s.

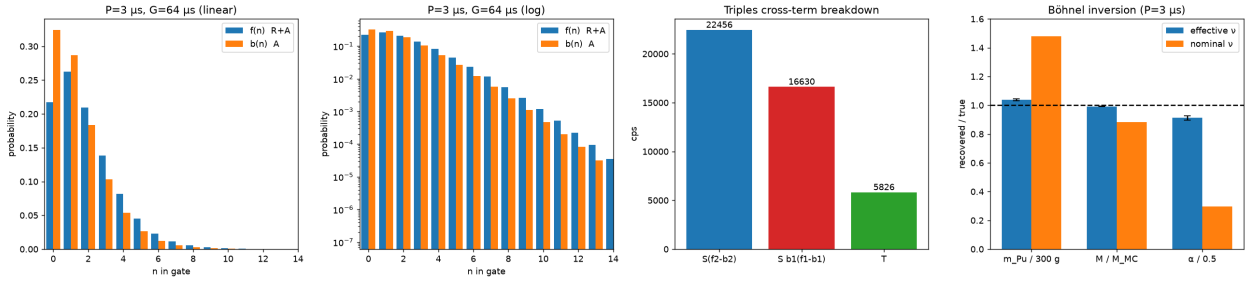


Figure 5: Shift register ($P = 3\,\mu$s). Left: R+A and A histograms. Third panel: the accidental cross-term $S\,b_1(f_1 - b_1)$ is about three times larger than the final Triples, so T_∞ is a small difference of large numbers. Right: inversion with the effective moments (blue) and with the nominal Pu-239 moments (orange, +48% mass, caused by the OpenMC implicit-fission approximation).

Method 2A: DFT deconvolution of the histograms. The R+A distribution is the accidental distribution convolved with the distribution $p(k)$ of true correlated partners: $f = p * b$. A Fourier transform turns the convolution into a product, so $p = \mathrm{IFFT}(\tilde{f}/\tilde{b})$. Then $D_\infty = S\,r_1/f_d$ and $T_\infty = S\,r_2/f_t$ with $r_1 = \sum k\,p(k)$ and $r_2 = \sum \binom{k}{2} p(k)$. The cross-term disappears automatically. The results are identical to the shift register's, as they must be, but the method still needs f_d and f_t.

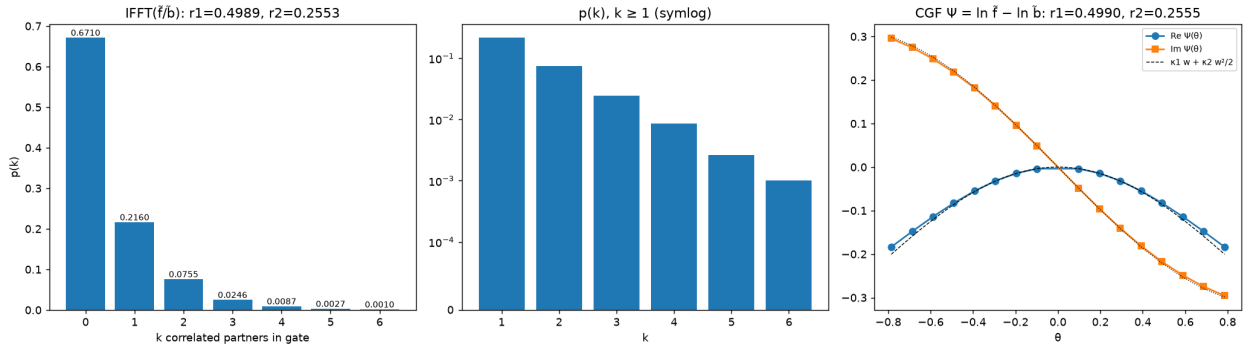


Figure 6: Method 2A. The deconvolved partner distribution $p(k)$, and the cumulant generating function $\Psi = \ln \tilde{f} - \ln \tilde{b}$, whose low-order terms give the same r_1, r_2.

4 New method: cross-spectrum and cross-bispectrum

4.1 Idea

Instead of gates, look at the pulse train as a signal. Bin each bank into $\Delta t = 2\,\mu$s bins, giving $x(t)$, $y(t)$, $z(t)$. Cut the data into segments of $T_{\rm seg} = 16.4$ ms (18 310 segments) and Fourier-transform each segment ($X(f)$, $Y(f)$, $Z(f)$). A correlated burst born at time t_0 adds a phase $e^{-i\omega t_0}$ to every neutron it produces. In a product of spectra in which the phases cancel, the burst survives averaging. Uncorrelated neutrons average to zero. The detector die-away enters only through its transfer function $h(\omega)$. For an exponential die-away $h(\omega) = 1/(1 + i\omega\tau)$, and in general $h(0) = 1$. This is the key point: **at zero frequency the die-away shape drops out**, so no gate fraction is needed.

4.2 Doubles from the cross power spectrum

The power spectrum of the total signal is

$$S_{ss}(f) = \frac{\langle |X + Y + Z|^2 \rangle}{T_{\rm seg}} = S + 2D_\infty |h(\omega)|^2. \tag{6}$$

The white floor is S, and the low-frequency plateau above it is $2D_\infty$. The **cross**-PSD between different banks removes the floor, because one neutron can only be in one bank:

$$S_{\rm cross}(f) = \frac{\mathrm{Re}\langle X^*Y + Y^*Z + Z^*X \rangle}{T_{\rm seg}} = 2D_\infty\, |h(\omega)|^2 \sum_{i<j} p_i p_j, \tag{7}$$

with p_i the bank detection fractions ($\sum_{i<j} p_i p_j = 1/3$ for equal banks).

4.3 Triples from the cross-bispectrum

The bispectrum is the next order. It multiplies three spectra whose frequencies add to zero:

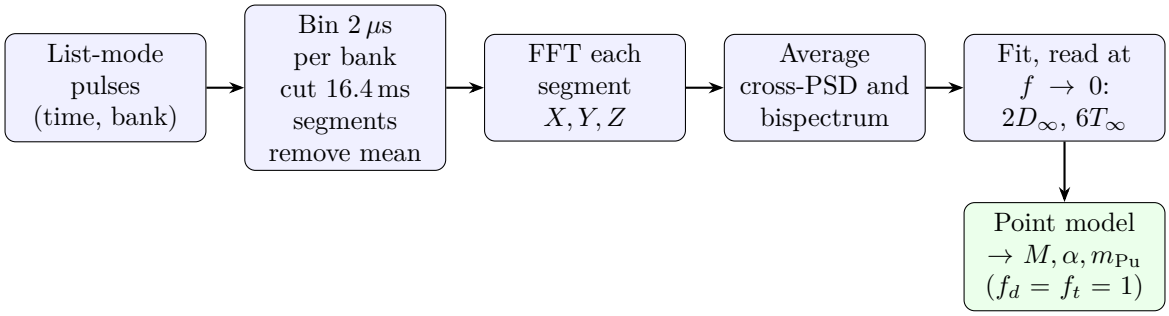
$$B_{xyz}(f_1, f_2) = \frac{1}{T_{\rm seg}} \mathrm{Re}\Big\langle \overline{X(f_1)\, Y(f_2)\, Z^*(f_1 + f_2)} \Big\rangle, \tag{8}$$

where the bar means an average over the 6 assignments of banks to the three slots. For a triplet born at t_0, the phases $e^{-i\omega_1 t_0} e^{-i\omega_2 t_0} e^{+i(\omega_1+\omega_2)t_0}$ cancel. This gives

$$\frac{B_{xyz}(f_1, f_2)}{p_X p_Y p_Z} = 6\, T_\infty\, \mathrm{Re}\big[h(\omega_1)\, h(\omega_2)\, h^*(\omega_1 + \omega_2)\big]. \tag{9}$$

Singles and Doubles vanish identically: a single neutron hits one bank and a pair at most two, but the product needs a neutron in each of the three banks. Only true triplets (or larger bursts) contribute. The background subtraction that dominates the shift-register Triples ($S\, b_1(f_1 - b_1)$) is simply not there.

4.4 How the method is applied



1. Bin and segment the pulse train. Subtract each segment's mean count, so the uncorrelated DC terms cannot leak into the products.

2. Accumulate the cross-PSD for $0 < f \leq 100\,\mathrm{kHz}$ and the bispectrum on the grid $f_1 \in (0, 10]\,\mathrm{kHz}$, $f_2 \in [-10, 10]\,\mathrm{kHz}$ (163×326 points).

3. Fit the spectra with a two-component die-away kernel $h(\omega) = \frac{1-w}{1+i\omega\tau_1} + \frac{w}{1+i\omega\tau_2}$ and read the zero-frequency amplitudes. The bispectrum gets its own free kernel.

4. Insert S (counted), D_∞ and T_∞ into the point model with $f_d = f_t = 1$.

5. Statistical errors come from the scatter over 10 independent time blocks.

The two-component kernel matters. The real chain timing is not a pure exponential, because some neutrons return from the moderator and cause a delayed thermal fission in the Pu. With the single exponential $1/(1+i\omega\tau)$, the fits extrapolate to values that are too low: $D_\infty = 17116$ (-1.7%) and $T_\infty = 13191$ (-4.7%). With the flexible kernel the zero-frequency values agree with the truth.

Check on a toy signal. Before the OpenMC data, the analysis was run on a synthetic 3-bank stream with known rates and a $45\,\mu$s exponential die-away. It gave $D_\infty = 7730$ (true 7800) and $T_\infty = 2690$ (true 2700). The cross-PSD floor came out -1.1 ± 4.3, consistent with zero.

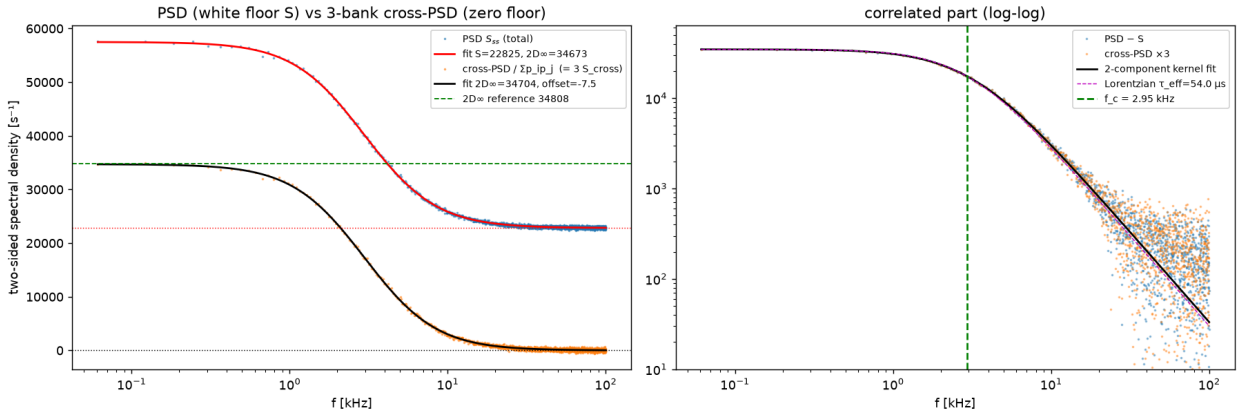


Figure 7: Doubles. Left: the total PSD sits on a white floor $S = 22825\,\mathrm{s}^{-1}$, while the 3-bank cross-PSD falls to zero (fitted offset -7.5 ± 4.9). Both plateaus give $2D_\infty \approx 34\,700$, against the reference 34 808. Right: the correlated part on a log scale; corner frequency $f_c \approx 3\,\mathrm{kHz}$.

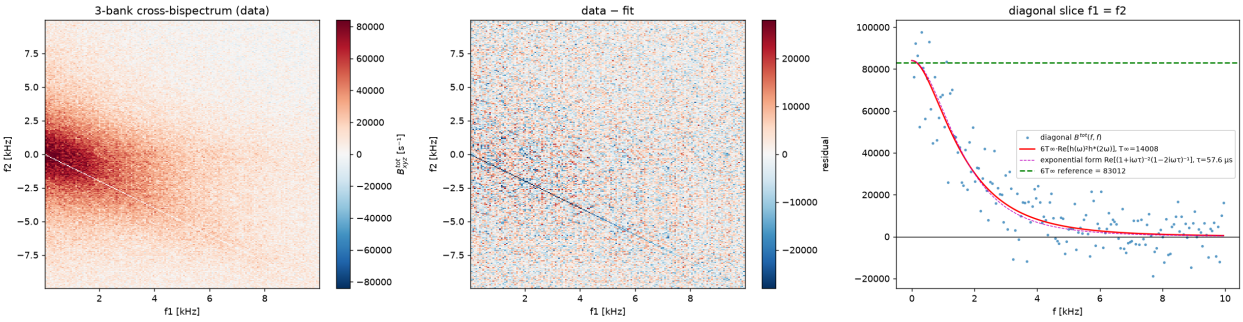


Figure 8: Triples. Left: measured 3-bank cross-bispectrum $B^{\mathrm{tot}} = B_{xyz}/(p_X p_Y p_Z)$. The signal sits near the origin and decays above a few kHz. The line $f_1 + f_2 = 0$ is excluded (zero after mean removal). Middle: residual after the fit, which is noise only. Right: the diagonal slice $f_1 = f_2$. The fitted curve reaches $6T_\infty$ at $f \to 0$, close to the reference (dashed).

5 Results

Quantity	Ground truth	Shift register	2A DFT	**New: cross-PSD + bispectrum**
S (s^{-1})	22854	22822 ± 12	22822 ± 12	22822 ± 12
D_∞ (s^{-1})	17404	17341 ± 40 (-0.4%)	same	17352 ± 75 (-0.3%)
T_∞ (s^{-1})	13835	13421 ± 78 ($\mathbf{-3.0}\%$)	same	14008 ± 532 ($+1.2\%$)
M	1.292	1.281 ± 0.002	same	1.299 ± 0.014
α	0.50	0.456 ± 0.008	same	0.53 ± 0.08
m_{eff} (g)	18.00	18.68 ± 0.12	same	17.5 ± 0.8
m_{Pu} (g)	**300**	$\mathbf{311.3 \pm 2.0}$ ($+3.8\%$)	same	$\mathbf{291 \pm 14}$ (-2.9%)

Table 2: Comparison for one 300 s measurement. Errors are 1σ statistical.

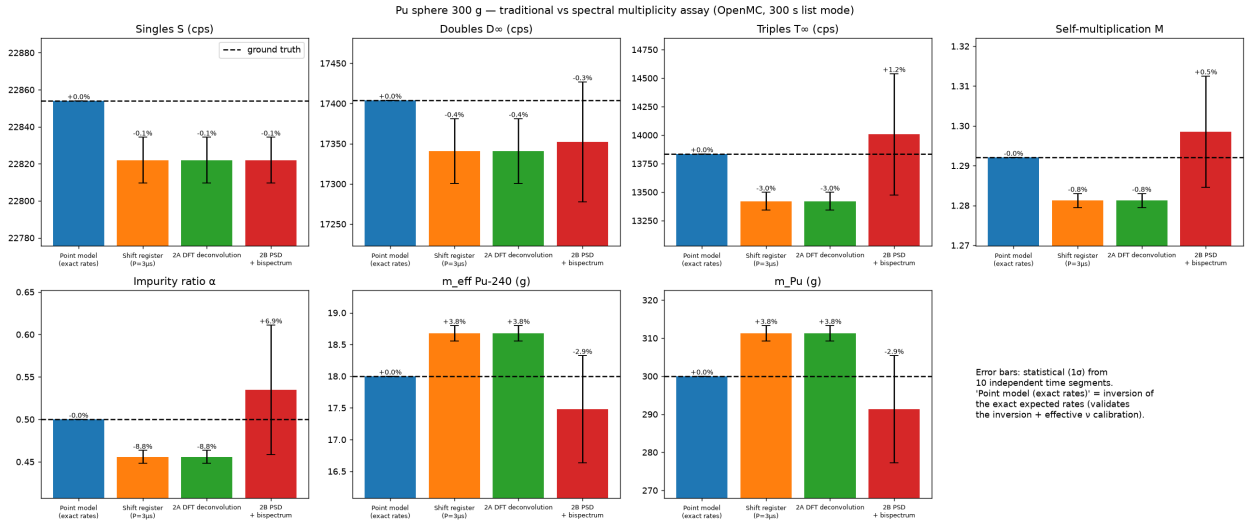


Figure 9: Recovered quantities against ground truth (dashed). "Point model (exact rates)" inverts the exact expected rates and returns exactly 300 g, which validates the inversion and the effective-moment calibration.

6 Why use the bispectrum? The case and its limits

6.1 Arguments for

1. **No gate fractions.** The shift register converts gated counts to rates with f_d and f_t. These depend on the die-away *and* on the time structure inside a fission chain. The spectral rates are read at $f \to 0$, where $h(0) = 1$ for any die-away shape. The kernel is fitted on every measurement, so the method calibrates itself.

2. **Background cancels by construction.** The cross-PSD has no Singles floor. The 3-bank bispectrum has neither a Singles nor a Doubles term. The shift-register Triples instead come from a subtraction whose cross-term $S\,b_1(f_1 - b_1)$ is about three times larger than the result.

3. **Diagnostic information.** The fitted kernel shows the real time structure of the detected chains, for example a slow component from neutrons that return to the item. A gate count cannot show this.

4. **Observed accuracy.** In this study the bispectrum T_∞ is $+1.2\%$ from truth (within 1σ). The shift-register T_∞ is -3.0% (about 5σ).

6.2 Arguments against (so far)

1. **Precision.** The bispectrum T_∞ is about 7 times noisier ($\pm 3.8\%$ against $\pm 0.6\%$), so the mass error is ± 14 g against ± 2 g. The total distance from truth is about the same for both methods (9 g against 11 g).

2. **The shift-register bias is partly our choice.** We computed f_t assuming the neutrons in a chain are detected independently. A real counter calibrates f_t with a Cf-252 source, which may remove part of the bias.

3. **Part of the non-exponential timing comes from the simulation.** OpenMC's implicit fission treatment changes how return-neutron fission chains look.

4. **One measurement.** We have one 300 s realisation, with no spread over repeated measurements.

5. **Hardware and cost.** At least three independent detector banks are needed, and the analysis takes minutes of computing instead of being real time.

6.3 Where a real case could be

The method should win where gate fractions are *not* transferable. This happens when a moderating or hydrogenous item changes the die-away, so a Cf calibration made in the empty well gives the wrong f_t, while the bispectrum adapts. Two other possible advantages are high (α, n) count rates, where the shift-register cross-term grows like S^2, and a *hybrid* method that uses the bispectrum kernel to correct f_t for the shift register. Section 7 tests these points.

7 Studies: testing the case

Four studies, all with the same detector, 300 s measurements and the analysis of Sections 3–4 (code in `pu_multiplicity_spectral/06_studies/`).

A **Repeated measurements**: 8 independent 300 s pulse trains of the baseline item. The spread across runs is the real 1σ error of one measurement.

B **Realistic calibration**: f_d, f_t calibrated like on a real counter, with a Cf-252 source (OpenMC run, empty cavity, known D_∞, T_∞). They are applied to the baseline item and to a **moderated item**: the same Pu sphere inside a 1.5 cm polyethylene shell (OpenMC run, $M = 1.326$, $\tau = 54.4\,\mu$s), 4 runs.

C (α, n) **scan**: $\alpha = 0$, 0.5, 2, 5 (Singles from 15 to 91 kcps).

D **Hybrid**: the shift register with $f_d = D/D_\infty^{\mathrm{PSD}}$ and $f_t = T/T_\infty^{\mathrm{bis}}$, calibrated on the *other* runs of the same item (leave-one-out), then applied to each run. A first attempt, computing f_t from the fitted bispectrum kernel shape, failed: it was 3% off.

What the studies show.

1. **The gate fraction does not transfer** (Fig. 10c). Cf-252 gives $f_t = 0.445$, while the true values are 0.427 (baseline) and 0.413 (moderated). A Cf fission is a single burst. A Pu item adds chain timing: neutrons return from the moderator and cause delayed fissions in Pu-239, which spreads the triplets in time, and a polyethylene shell makes this stronger. Triples are therefore underestimated by 4% and 7%. This is the classic matrix effect. The calibration source cannot see it, but the bispectrum measures it directly.

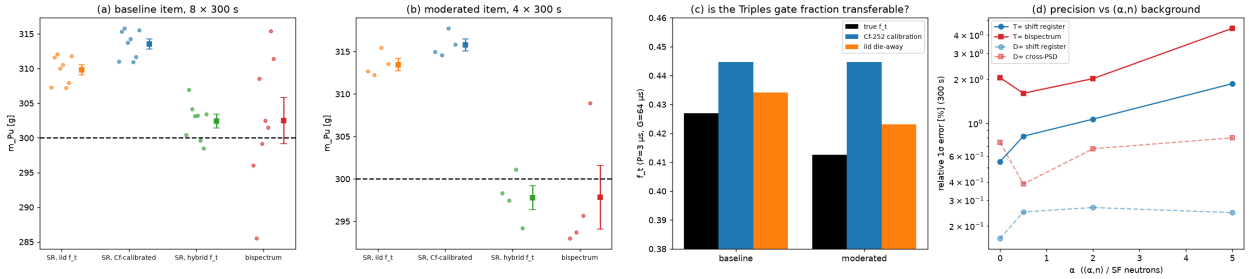


Figure 10: (a, b) Mass per run (dots) and mean ± standard error (squares) for four estimators, baseline and moderated item. (c) Triples gate fraction: the true value (T/T_∞) compared with the Cf-252 calibration and the iid die-away estimate. (d) Relative 1σ error of D_∞ and T_∞ against the (α, n) ratio, from the in-run block errors; these overestimate the bispectrum error by about 1.5× (Study A).

Item	Method	mean m_{Pu} (g)	bias	spread / run (g)	RMS error (g)
Baseline (8 runs)	SR, iid f_t	309.8	+3.3%	2.1	10.0
	SR, Cf-252 calibrated	313.5	+4.5%	2.1	13.7
	Bispectrum	302.5	$+0.8 \pm 1.1\%$	9.5	9.2
	Hybrid	302.4	$+0.8 \pm 0.3\%$	2.8	**3.5**
Moderated (4 runs)	SR, iid f_t	313.5	+4.5%	1.4	13.5
	SR, Cf-252 calibrated	315.8	+5.3%	1.4	15.8
	Bispectrum	297.8	$-0.7 \pm 1.3\%$	7.5	6.8
	Hybrid	297.8	$-0.7 \pm 0.5\%$	2.8	**3.3**

Table 3: Study results (300 s runs, true mass 300 g). RMS error = $\sqrt{\langle (m - 300)^2 \rangle}$ combines bias and spread.

2. **The bispectrum is unbiased on both items**: T_∞ is −0.6% from truth in both cases, with a real spread of 1.9% per run. The block-error estimate used in Table 2 (±3.8%) was pessimistic.

3. **On total error for a single measurement, the bispectrum already beats the calibrated shift register**: RMS 9.2 against 13.7 g, and 6.8 against 15.8 g for the moderated item.

4. **The hybrid is best**: shift-register precision without the bias, RMS about 3 g on both items. It needs a set of measurements of the same item type to calibrate f_t (here 7 other runs, about 35 min). No calibration source is required.

5. **No high-rate advantage** (Fig. 10d). The bispectrum stays 2–4 times less precise on T_∞ across $\alpha = 0$–5. Its noise grows with the Singles rate at least as fast as the shift-register cross-term does.

Caveat. The size of the chain-timing effect depends on how often neutrons return to the item and fission. OpenMC gets the mean fission rates right but treats fission implicitly, so the exact f_t shift in a real item may differ. The direction of the effect is physical: real Pu items do multiply returned thermal neutrons. It should be checked with an analog multiplicity code (e.g. MCNPX-PoliMi) or with measurements.

8 Conclusions

- **Shift register (legacy):** precise (≈ 2 g per 300 s run) but *biased* by its Triples gate fraction f_t. Calibrating f_t with Cf-252, as a real counter does, makes the bias *worse* (+4.5%, and +5.3% for a moderated item), because a Cf source has no fission-chain timing.

- **Cross-PSD and cross-bispectrum (new):** *unbiased* on both items, with no gate fractions and no calibration source. Its rates are read at zero frequency, where the die-away response equals one, and the background cancels by construction. It costs precision: about 9 g per 300 s run, 2–4 times worse on T_∞, with no gain at high (α, n) rates.
- **The case for the bispectrum** is *accuracy without calibration*, most of all for items whose matrix changes the chain timing. Even alone, its total error for a single measurement is smaller than that of a Cf-calibrated shift register.
- **Recommended use:** the **hybrid**. The bispectrum measures f_t for the item type, and the shift register delivers the precision. This gives about 3 g RMS on 300 g (1%).
- **Next steps:** confirm the chain-timing effect with an analog fission code or with measurements; improve the bispectrum precision (frequency weighting, constrained kernel); test a single-measurement hybrid.
- **Model limitation:** OpenMC fixed-source fission is implicit. The absolute induced-fission moments are effective values, but all methods analyse the same pulse trains, so the comparison is valid.

Reproducibility. All code is in `pu_multiplicity_spectral/`, organised as one folder per step (00–06), each with a smoke test, logs and plots. The geometry figures are made by `documentation/make_geometry_figures.py`.